

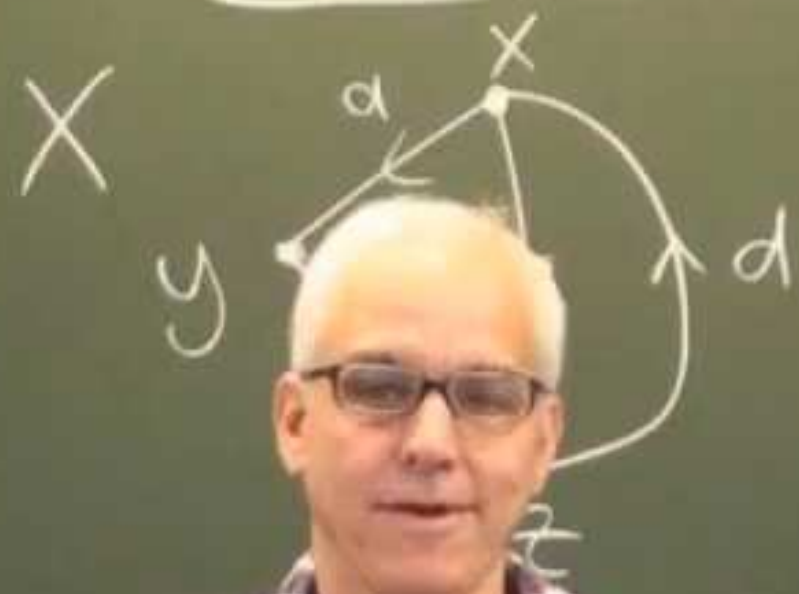
Algebraic Topology: a beginner's course

This lecture:

**AlgTop31: An introduction
to homology (cont.)**

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Lecture 31 Introduction to Homology (cont.)



$$\partial(a) = y - x$$

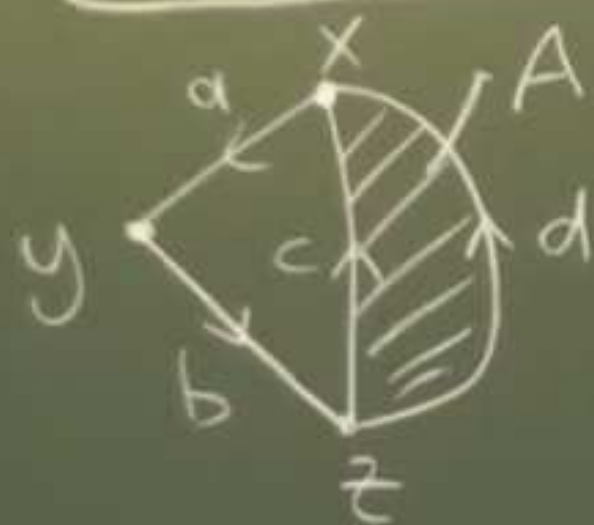
$$\partial(b) = z - y$$

$$\partial(c) = x - z$$

$$\partial(d) = x - z$$

Lecture 31 Introduction to Homology (cont.)

Now modify $X \rightarrow X_2$ by adding a 2-dim cell A between edges c, d .



$$\partial(a) = y - x$$

$$\partial(b) = z - y$$

$$\partial(c) = x - z$$

$$\partial(d) = x - z$$

$$\partial: C_1 \rightarrow C_0$$

A cycle in C_1 is $t \in C_1$ with $\partial(t) = 0$.

Cycles: spanned by $a+b+c, a+b+d$

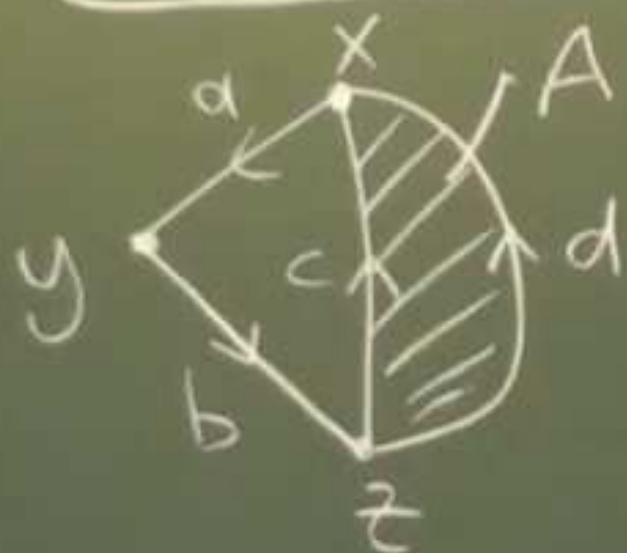
C_0 : formal integer combinations of x, y, z : $3x - 5y + \dots$
(0-dim chains)

C_1 : formal integer combinations of a, b, c, d ie $3a - \dots$
(1-dim chains)

$$H_1(X) = \text{'cycles'} = \mathbb{Z} + \mathbb{Z}$$

Lecture 31 Introduction to Homology (cont.)

Now modify $X \rightarrow X_2$ by adding a 2-dim cell A between edges c, d .
orient A clockwise



$$\partial(a) = y - x$$

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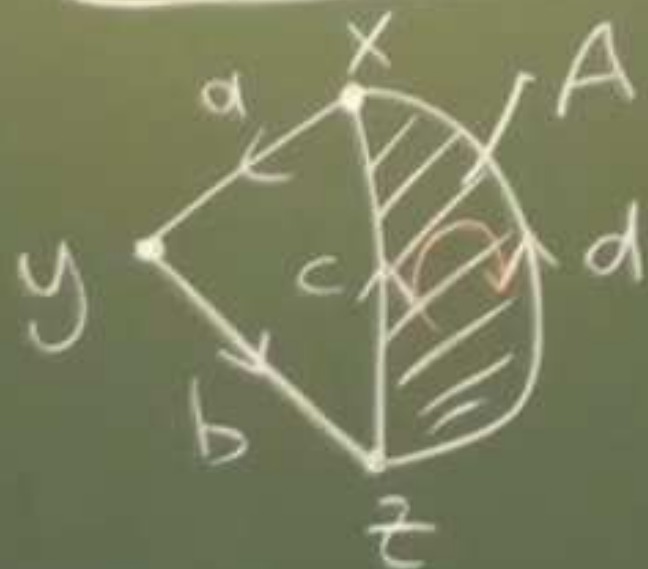
C_0 : formal integers x, y, z : $3x -$
(0-dim chain)

C_1 : formal integers a, b, c, d i.e.
(1-dim chains)

$$H_1(X) = \text{'cycles'} = \mathbb{Z}$$

Lecture 31 Introduction to Homology (cont.)

Now modify
 $X \rightarrow X_2$ by
 adding a
 2-dim cell
 A between
 edges c, d
 orient A
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$$\partial(a) = y - x$$

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$$\partial: C_1 \rightarrow C_0$$

A cycle in C_1 is $t \in C_1$ with $\partial(t) = 0$.

Cycles: spanned by $a+b+c, a+b+d$

C_0 : formal int

x, y, z :

(0-dim ch)

C_1 : formal in

a, b, c, d

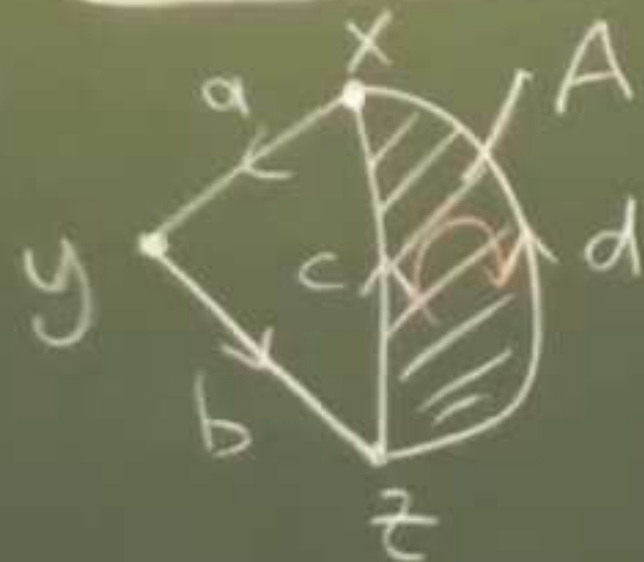
(1-dim ch)

$H_1(X) = \text{'cycles'}$

Lecture 31 Introduction to Homology (cont.)

Now modify
 $X \rightarrow X_2$ by
 adding a
 2-dim cell

X



$$\partial(a) = y - x$$

$$\partial(b) = z - y$$

$$\partial(c) = x - z$$

$$\partial(d) = x - z$$

$$\partial: C_1 \rightarrow C_0$$

A cycle in C_1 is $t \in C_1$ with $\partial(t) = 0$

Cycles: spanned by $a+b+c, a+b+d$

So $\partial(A) = c - d$

The cycle $c-d$ is now the

C_0 : formal integ

$x, y, z: 3x$

(chain)

C_1 : formal inte

d is

chain

C_2 : formal integer combs of
 $A: 3A, -5A$
 (2-dim chains).

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

$$\partial_1: C_1 \rightarrow C_0$$

$$\partial_2: C_2 \rightarrow C_1$$

$$H_1(X_2) \simeq \mathbb{Z} + \mathbb{Z}$$

$$\simeq \mathbb{Z}$$

$$\mathbb{Z}$$

$$\leftarrow \langle c-d \rangle$$

$$\leftarrow \langle atb+c, atb+ds \rangle$$

The presence of the 2-cell A
 that the cycle $c-d$ is now
 to a constant (0): no longer

Algebraically: want $c-d$

But then $atb+c = atb+$

We need to quotient the
 group of ^{1-dim} cycles (gen'd by atb)
 subgroup of 1-dim boundaries (ge

$$\mathbb{Z} / B_1$$

$$Z_1 \equiv 1\text{-dim cy}$$

$$B_1 \equiv 1\text{-dim bo}$$

roduction to H

$$(a) = y - x$$

$$(b) = z - y$$

$$(c) = x - z$$

$$(d) = x - z$$

$$\Rightarrow C_0$$

$$\Rightarrow C_1$$

$$\Rightarrow C_2$$

$$\Rightarrow C_3$$

$$\Rightarrow C_4$$

$$\Rightarrow C_5$$

$$\Rightarrow C_6$$

$$\Rightarrow C_7$$

$$\Rightarrow C_8$$

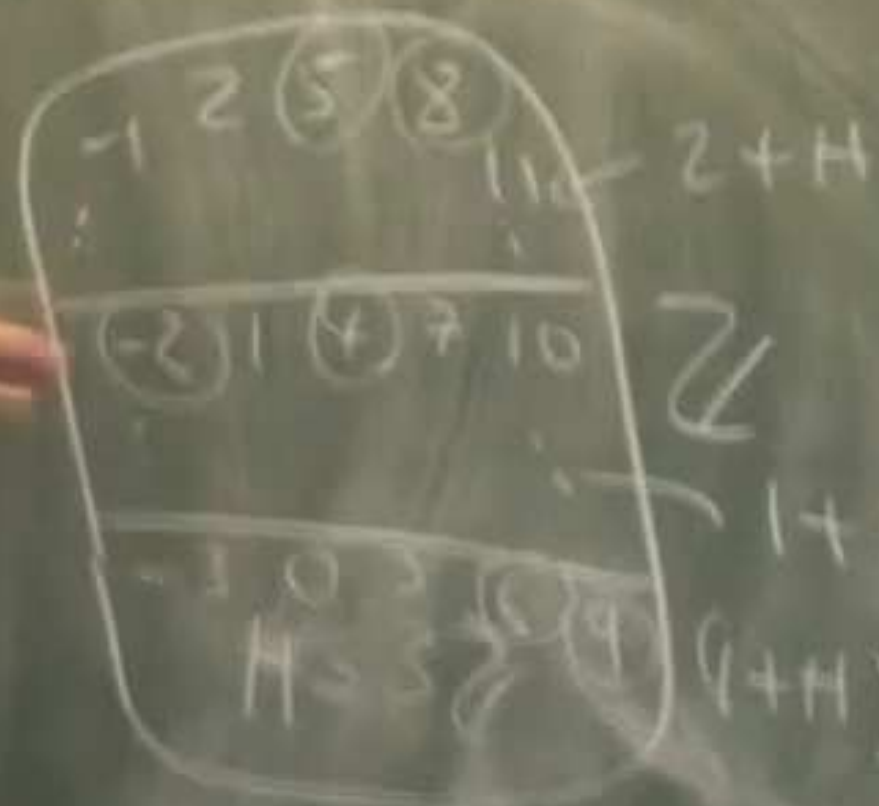
Quotients

$$\mathbb{Z}/3\mathbb{Z}$$

$$H = 3\mathbb{Z} = \dots, -3, 0, 3, 6, \dots$$

$$\mathbb{Z} = \dots, -1, 0, 1, 2, 3, \dots$$

$\mathbb{Z}/H$: cosets
of
 H in $\mathbb{Z}$



$$\mathbb{Z}$$

$\Rightarrow$ mod 3
arithmetic

C_2 : formal integers

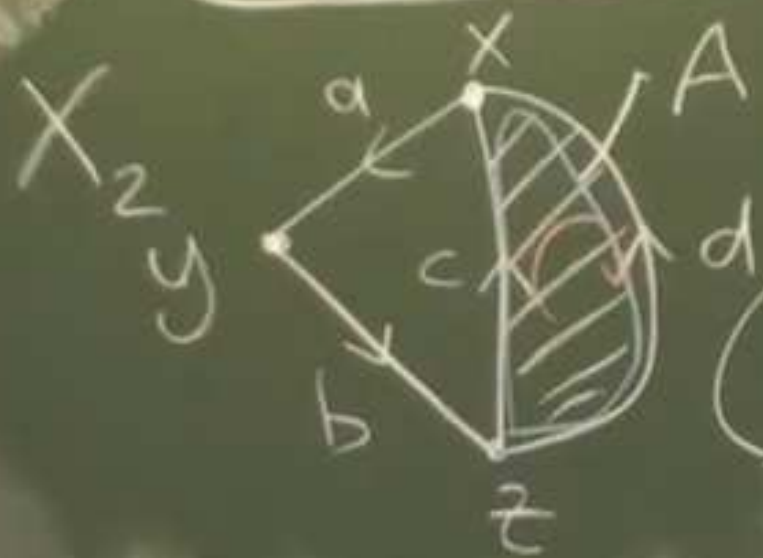
$$A: 3A,$$

(2-dim char)

$$C_2 \xrightarrow{\partial_2} C_1$$

$$H_1(X_2) \approx \mathbb{Z} \oplus \mathbb{Z} \approx \mathbb{Z}^2$$

Lecture 31 Introduction to H



$Z_1 = \text{cycles.}$

$B_1 \rightarrow$

$a+b+c$		
$a+b+d$		
$c+d$		

$a+b+c, a+b+d$

Quotients $\mathbb{Z}/3\mathbb{Z}$

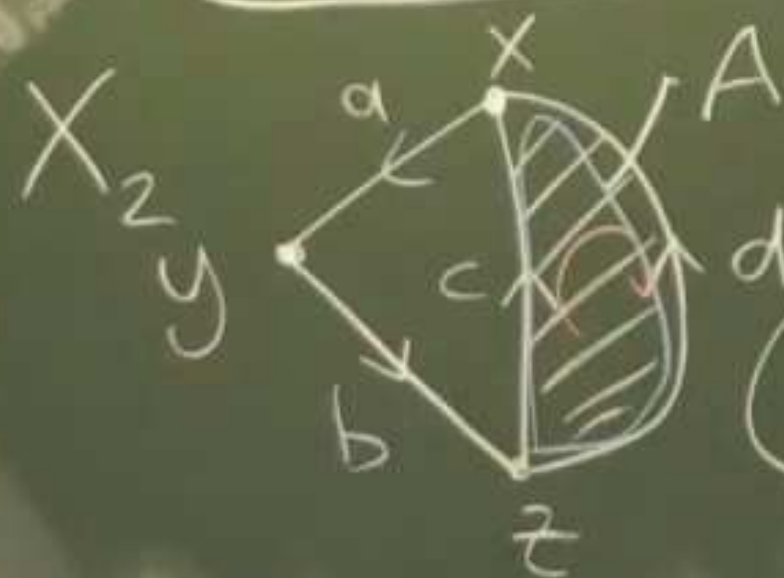
$H = 3\mathbb{Z} = \dots, -3, 0, 3, 6, \dots$
 $\mathbb{Z} = \dots, -1, 0, 1, 2, 3, \dots$

-1	2	5	8	$11 \leftarrow 2+H$
-2	1	4	7	10
-3	0	3	6	9

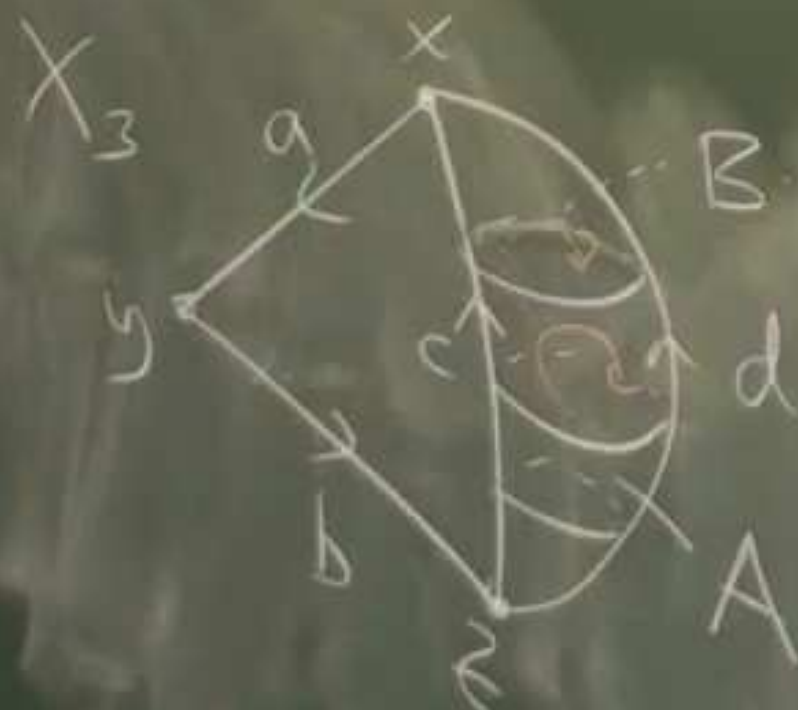
$H = 3\mathbb{Z}$

Introduce another Lecture 31 Introduction to H

2-cell B
between
 $c+d$



$Z_1 = \text{cycles}$

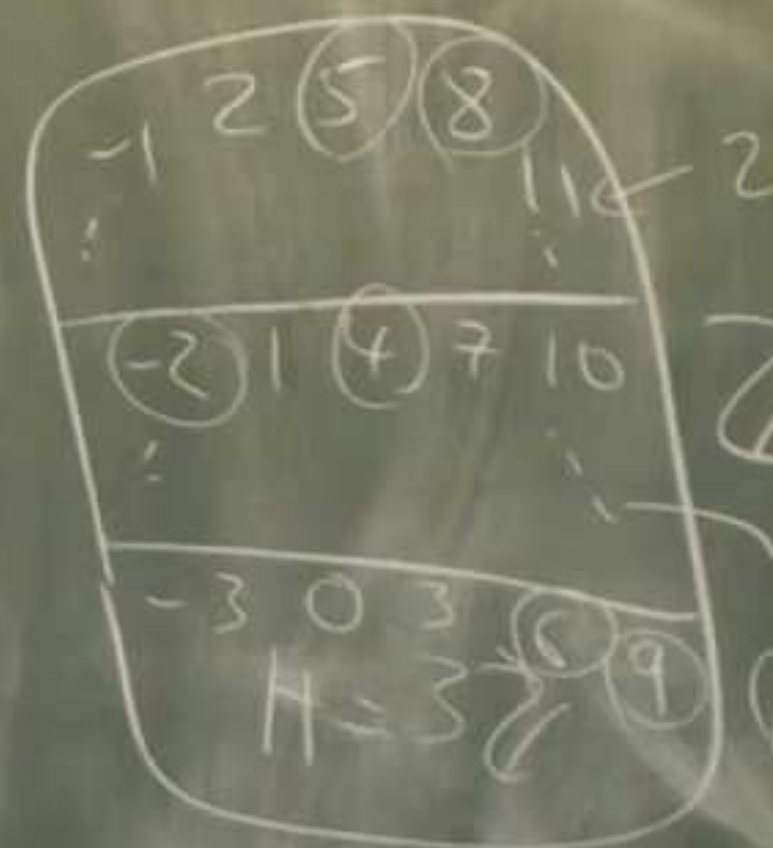


$$\partial(B) = c - d = \partial(A)$$

Quotients $\mathbb{Z}/$

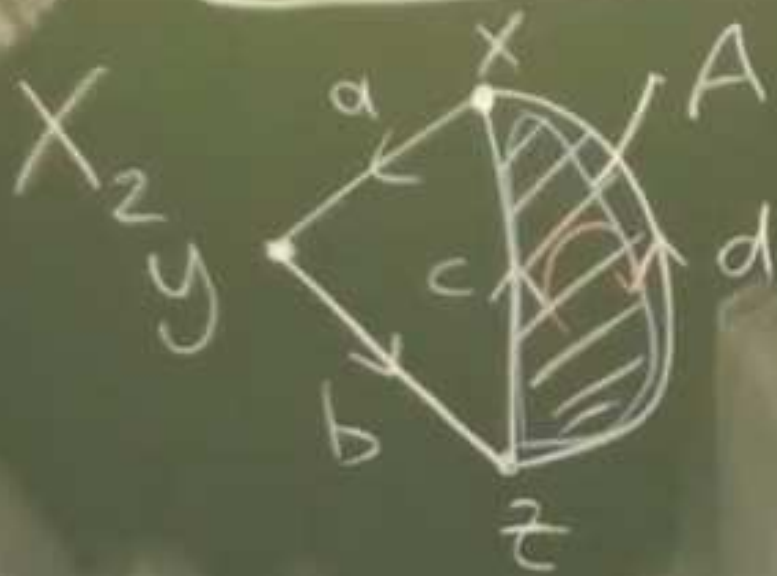
$$H = 3\mathbb{Z} = \dots, -3, 0, 3, \dots$$

$$\mathbb{Z} = \dots, -1, 0, 1, 2, \dots$$



Introduce another Lecture 31

2-cell B
between
 $c+d$

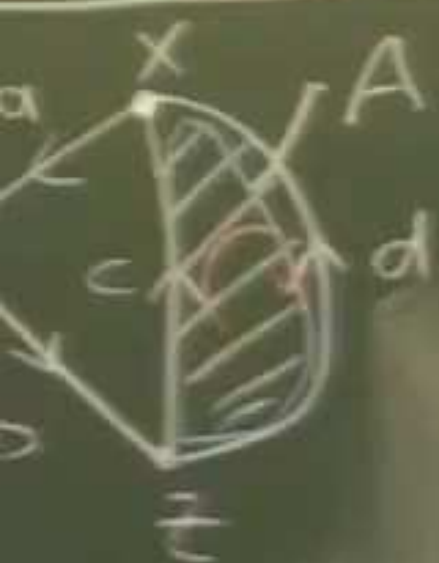


$$\partial(B) = c - d \\ = \partial(A)$$

Now with X_3 : $C_2 = \langle A, B \rangle$

d

ecture 31



Now with X_3 : $C_2 = \langle A, B \rangle$ int. linear combo
eg. $3A - 4B$.

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

$$\partial_2(A) = c - d = \partial_2(B)$$

$$H_2(X_3) = \langle A - B \rangle \cong \mathbb{Z}$$

$$H_1(X_3) = \mathbb{Z}_1 / B_1 = \frac{\ker \partial_1}{\text{im } \partial_2} = \frac{\mathbb{Z} + \mathbb{Z}}{\mathbb{Z}} \cong \mathbb{Z}$$

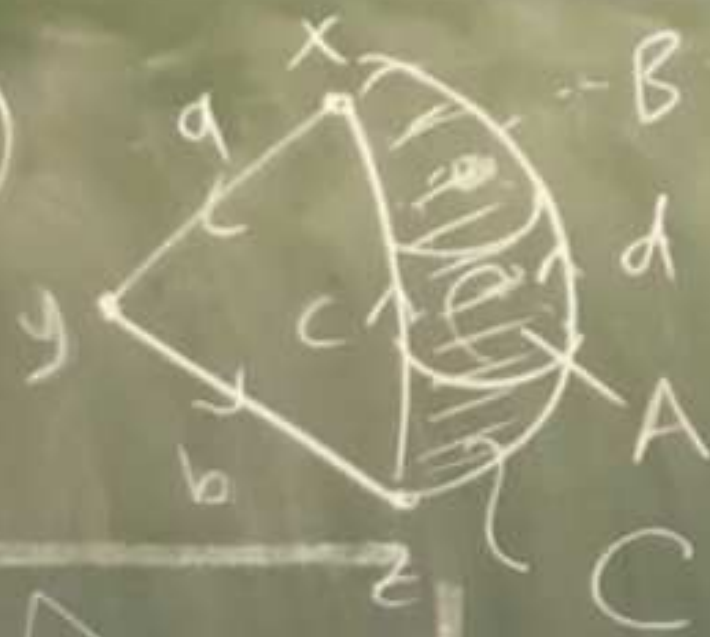
same as $H_1(X_2)$.

Now ∂_2 has a non-trivial kernel, $\ker \partial_2$

$$\partial(B) = c - d = \partial(A)$$

$A - B$ is a $\mathbb{Z}$ -dim cycle, generating second homology

X_4



Let's attach a 3-cell (ball in 3-dim space) C along the 2-sphere formed by $A + B$.

$$\begin{array}{ccccccc} C_3 & \xrightarrow{\partial_3} & C_2 & \xrightarrow{\partial_2} & C_1 & \xrightarrow{\partial_1} & C_0 \\ | & & | & & | & & | \\ \langle C \rangle & & \langle A, B \rangle & & \langle a, b, c, d \rangle & & \langle x, y, z \rangle \end{array}$$

Orient C so $\partial(C) = A - B$ (this ensures that



Let's attach a 3-cell (ball in 3-dim space) C along the 2-sphere formed by $A \pm B$.

$$\begin{array}{ccccccc}
 C_3 & \xrightarrow{\partial_3} & C_2 & \xrightarrow{\partial_2} & C_1 & \xrightarrow{\partial_1} & C_0 \\
 | & & | & & | & & | \\
 \langle C \rangle & & \langle A, B \rangle & & \langle a, b, c, d \rangle & & \langle x, y, z \rangle
 \end{array}$$

Orient C so $\partial(C) = A - B$ (this ensures the boundary is a cycle)